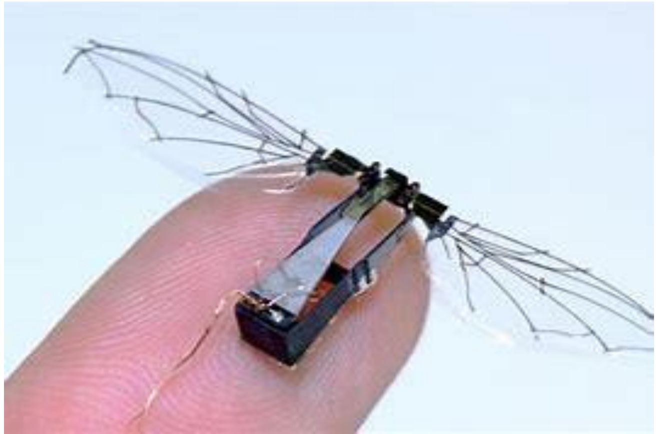
TYPES OF ROBOTS

Types of Robots

- Mechanical bots come in all shapes and sizes to efficiently carry out the task for which they are designed
- All robots vary in **design, functionality** and **degree of autonomy**.
- From the 0.2 millimeter-long “RoboBee” to the 200 meter-long robotic shipping vessel “Vindskip,” robots are emerging to carry out tasks that humans simply can’t.



Types of Robots (cont)

- Generally, there are five types of robots:
  - Pre-Programmed Robots
  - Humanoid Robots
  - Autonomous Robots
  - Teleoperated Robots
  - Augmenting Robots

Pre-Programmed Robots

- Pre-programmed robots operate in a controlled environment where they do simple, monotonous tasks.
- An example of a pre-programmed robot would be a mechanical arm on an automotive assembly line.
- The arm serves one function — to weld a door on, to insert a certain part into the engine, etc. — and its job is to perform that task longer, faster and more efficiently than a human.



Humanoid Robots

- Humanoid robots are robots that look like and/or mimic human behavior
- Usually perform human-like activities (*like running, jumping and carrying objects*), and are sometimes designed to look like us, even having human faces and expressions.
- Two most prominent examples of humanoid robots are **Hanson Robotics' Sophia** and **Boston Dynamics**' **Atlas**.
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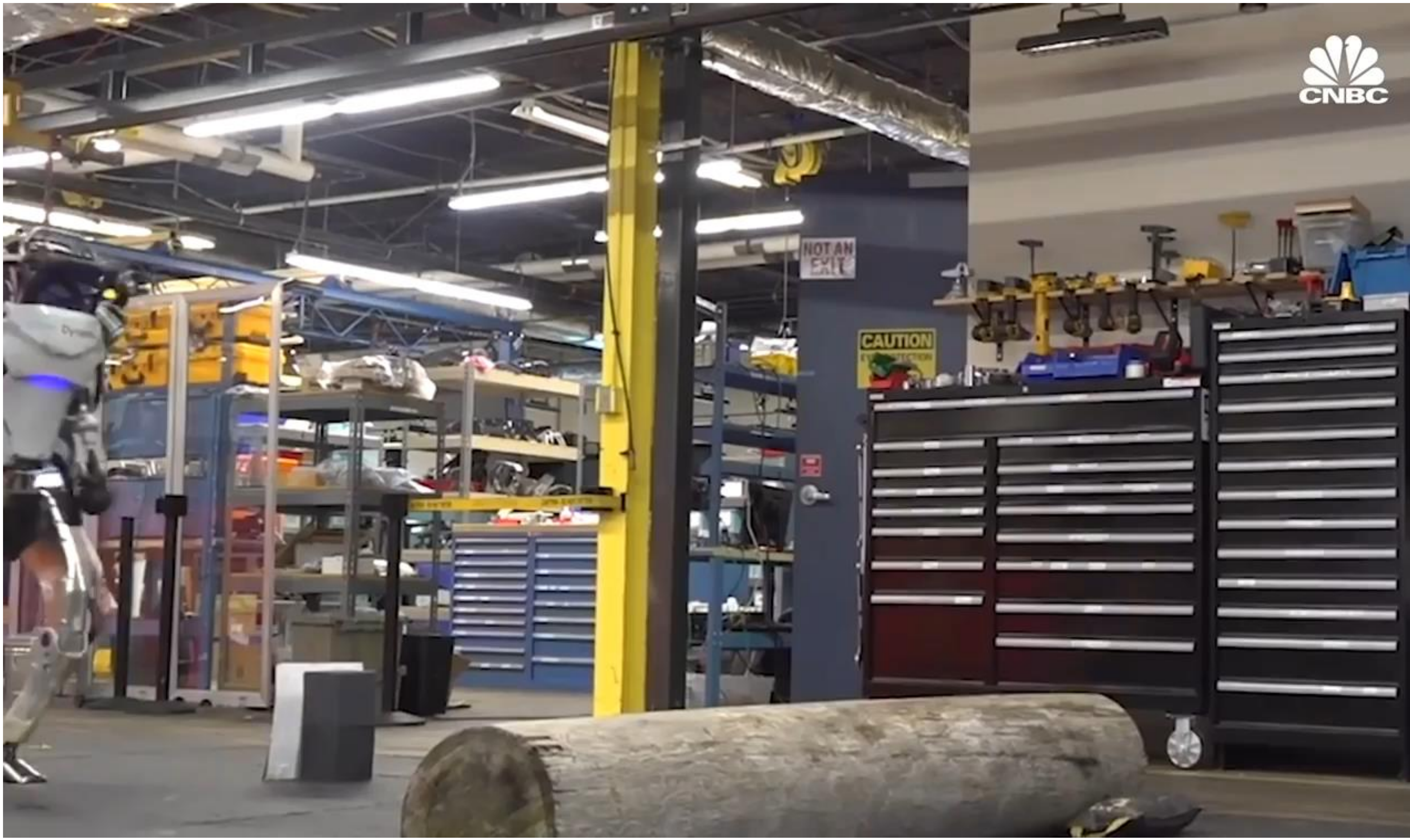
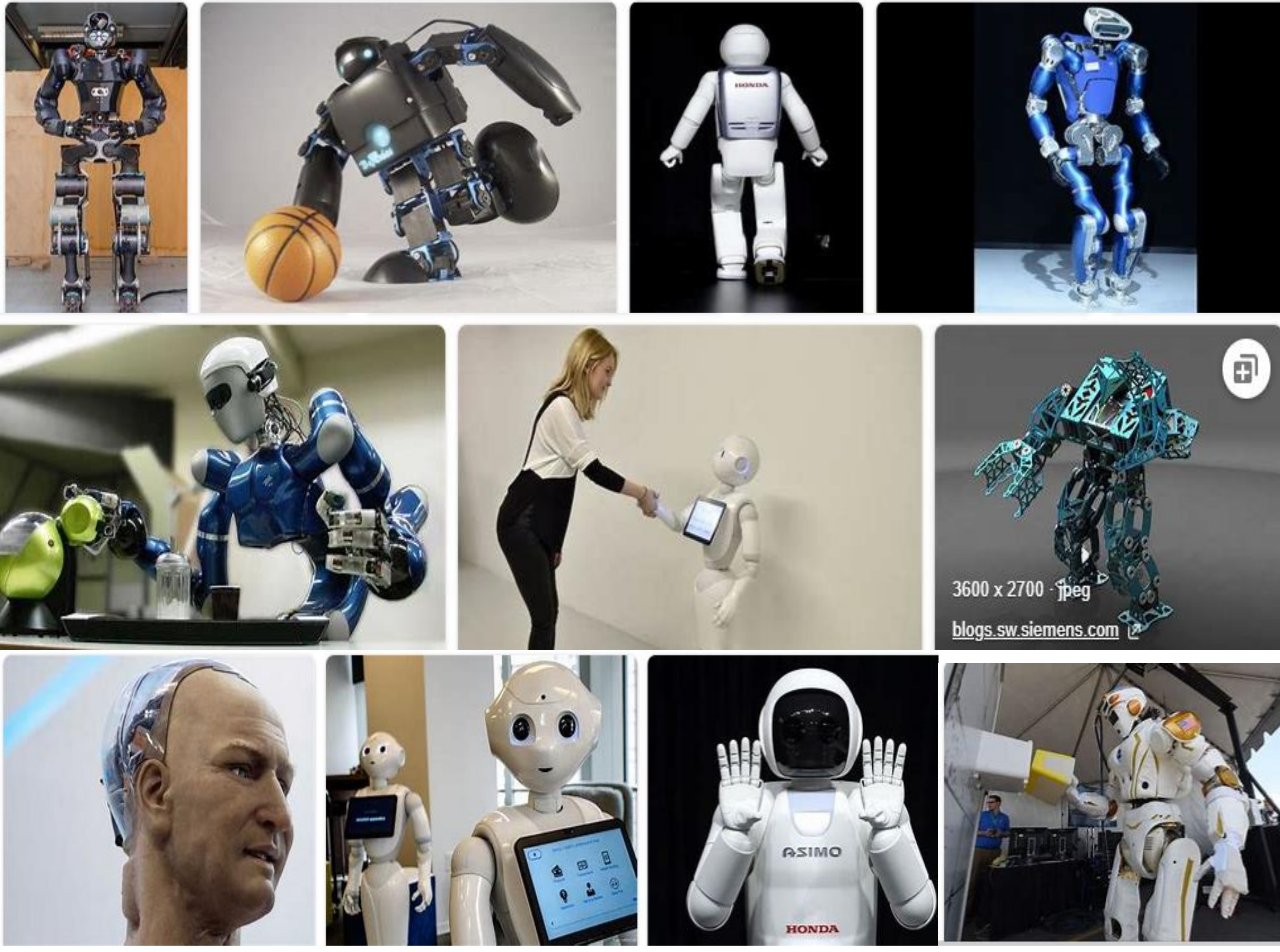
Sophia

Boston Dynamics Atlas

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Autonomous Robots

- Autonomous robots operate independently of human operators
- Usually designed to carry out tasks in open environments that do not require human supervision
- They are quite unique because they use **sensors** to **perceive** the world around them, and then employ **decision-making** structures to take the optimal next step based on their data and mission

Examples of Autonomous robots

- Autonomous/Unmanned Ground Vehicles
  - Cleaning Bots (for example, Roomba)
  - Lawn Trimming Bots
  - Hospitality Bots
- Autonomous Aerial Vehicles (UAV)/Autonomous Drones
- Unmanned Surface Vehicles
- Unmanned Underwater vehicles

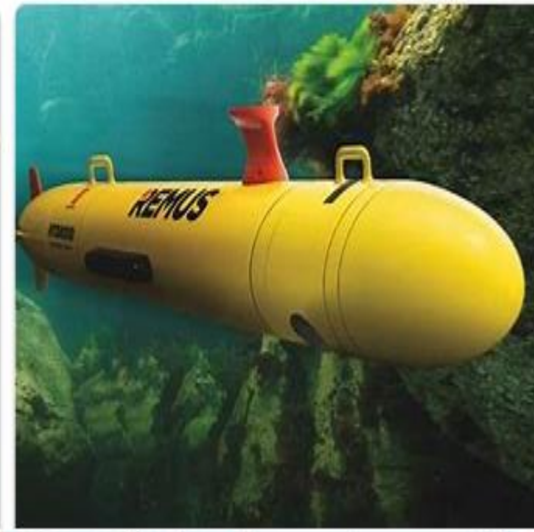
Autonomous/Unmanned Ground vehicles

Unmanned Aerial Vehicles (UAV)





Unmanned Underwater Vehicles (UUV)



Unmanned Surface Vehicle (USV)

Medical robots



Teleoperated Robots

- Teleoperated robots are semi-autonomous bots that use a wireless network to enable human control from a safe distance
- These robots usually work in extreme geographical conditions, weather, circumstances, etc.
- Examples :
  - Human-controlled submarines used to fix underwater pipe leaks during the BP oil spill
  - Drones used to detect landmines on a battlefield.
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Augmenting Robots

- They either **enhance** current human capabilities or **replace** the capabilities a human may have lost
- Provide the ability to redefine the definition of humanity by making humans faster and stronger
- **Examples--** robotic prosthetic limbs or exoskeletons used to lift hefty weights.
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Robotic prosthetic limbs or exoskeletons

















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Is robot software considered robotics?

- A software robot is an abundant type of computer program which carries out tasks autonomously, such as a chatbot or a web crawler.
- However, because software robots only exist on the internet and originate within a computer, they are not considered robots.
- In order to be considered a robot, a device must have a physical form, such as a body or a chassis.

END